

DW03_lfsr_load

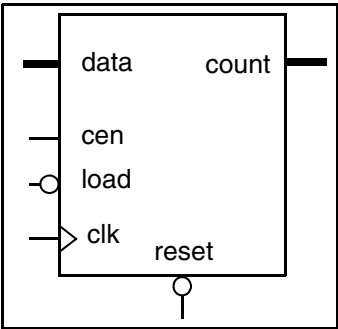
LFSR Counter with Loadable Input

Version, STAR, and myDesignWare Subscriptions: [IP Directory](#)

Features and Benefits

- Parameterized word length
- Loadable counter registers
- High speed, area-efficient
- Asynchronous reset
- Terminal count

Revision History



Description

DW03_lfsr_load is a parameterized word-length up counter with loadable data input. DW03_lfsr_load implements a counter as LFSR (linear feedback shift register) which also acts as a pseudo-random counter constructed as primitive characteristic polynomials.

Table 1-1 Pin Description

Pin Name	Width	Direction	Function
data	<i>width</i> bits	Input	Input data
load	1 bit	Input	Input load data to counter, active low
cen	1 bit	Input	Input count enable
clk	1 bit	Input	Clock
reset	1 bit	Input	Asynchronous reset, active low
count	<i>width</i> bits	Output	Output count bus

Table 1-2 Parameter Description

Parameter	Values ^a	Description
width	1 to 50	Word length of counter

a. The upper bound of the legal range is a guideline to ensure reasonable compile times.

Table 1-3 Synthesis Implementations

Implementation Name	Function	License Feature Required
str	Synthesis model	DesignWare

Table 1-4 Simulation Models

Model	Function
DW03.DW03_LFSR_LOAD_CFG_SIM	Design unit name for VHDL simulation
dw/dw03/src/DW03_lfsr_load_sim.vhd	VHDL simulation model source code
dw/sim_ver/DW03_lfsr_load.v	Verilog simulation model source code

Table 1-5 Counter Operation Truth Table

reset	load	cen	Operation
0	X	X	Reset
1	0	1	Data is XORd with count
1	X	0	Standby
1	1	1	Count up

The shift register is fed back from two or more taps. The number of taps does not increase with a large counter *width*.

An LFSR counter runs faster than a binary counter in synchronous systems because the first bit is calculated and the remaining bits are shifted.

DW03_lfsr_load can be used in built-in test circuitry for VLSI chips and multiple-input signature registers.

Counter Function

The data input bus, *data*, ranges from *width* - 1 to 0.

When the input signal, *load*, is low, *data* is exclusive ORd with *count*. When *load* is high, the counter operates as a normal LFSR up counter.

When the count enable pin, *cen*, is high, the counter is active. When *cen* is low, the counter is disabled and *count* remains at the same value.

The *reset* signal is an asynchronous reset that is active low. When *reset* is low, the counter output is "00...00". When *reset* is high, the counter operates normally.

The *count* is the output port of pseudo-random binary sequences, ranging from *width* - 1 to 0. For a list of the primitive polynomials, see the [Logic-Sequential Overview](#). A value of $2^{width-2}$ ("11...11") is an illegal state; therefore, the counter stops at "11...11".

Timing Diagrams

Figure 1-1 and Figure 1-2 show various timing conditions for DW03_lfsr_load.

Figure 1-1 Functional Operation: reset, count and count_enable Sequence

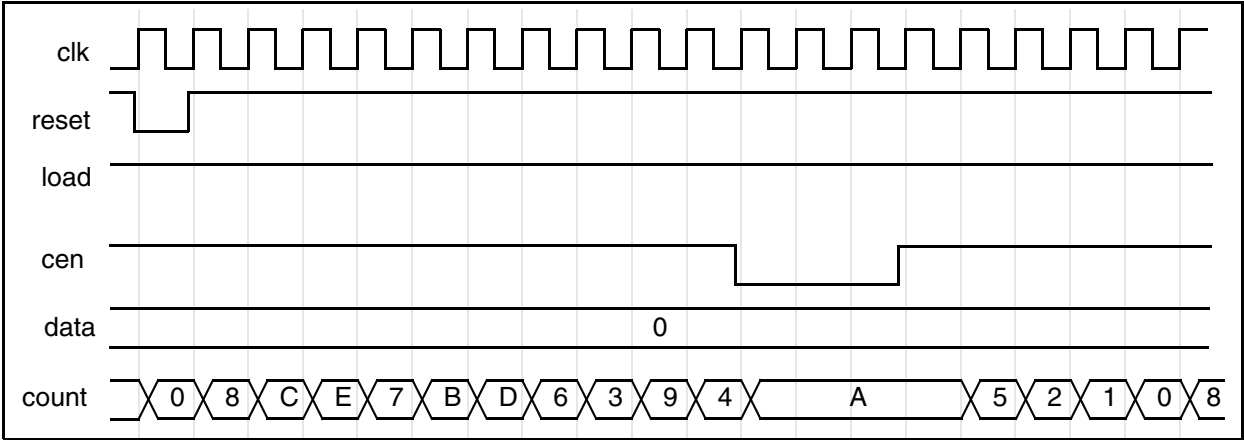
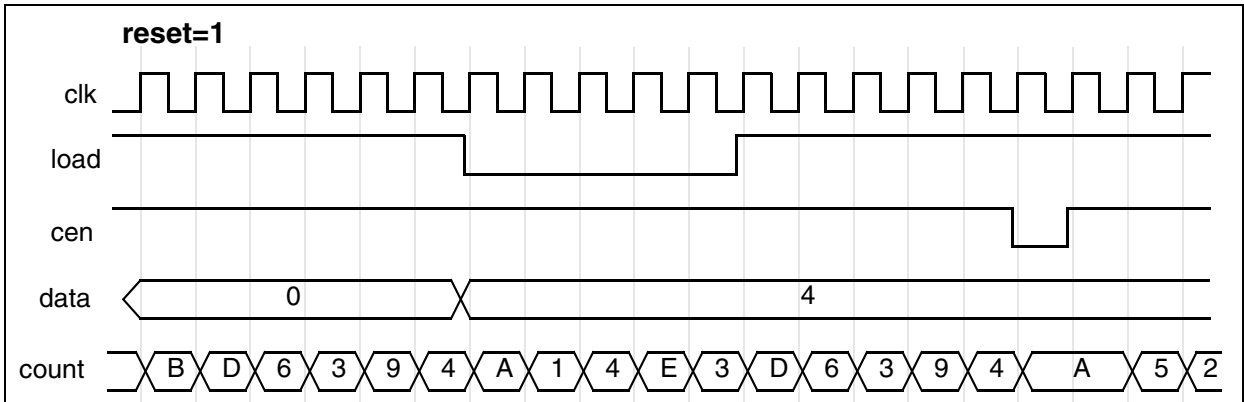


Figure 1-2 Functional Operation: load and count_enable Sequence



Related Topics

- [Logic – Sequential Overview](#)
- [DesignWare Building Block IP User Guide](#)

HDL Usage Through Component Instantiation - VHDL

```
library IEEE,DWARE;
use IEEE.std_logic_1164.all;
use DWARE.DWpackages.all;
use DWARE.DW_foundation_comp.all;

entity DW03_lfsr_load_inst is
  generic ( inst_width : INTEGER := 8 );
  port ( inst_data   : in std_logic_vector(inst_width-1 downto 0);
        inst_load    : in std_logic;
        inst_cen     : in std_logic;
        inst_clk     : in std_logic;
        inst_reset   : in std_logic;
        count_inst  : out std_logic_vector(inst_width-1 downto 0) );
end DW03_lfsr_load_inst;

architecture inst of DW03_lfsr_load_inst is
begin

  -- Instance of DW03_lfsr_load
  U1 : DW03_lfsr_load
    generic map ( width => inst_width )
    port map ( data => inst_data, load => inst_load, cen => inst_cen,
              clk => inst_clk, reset => inst_reset, count => count_inst );
end inst;

-- pragma translate_off
configuration DW03_lfsr_load_inst_cfg_inst of DW03_lfsr_load_inst is
  for inst
  end for; -- inst
end DW03_lfsr_load_inst_cfg_inst;
-- pragma translate_on
```

HDL Usage Through Component Instantiation - Verilog

```
module DW03_lfsr_load_inst( inst_data, inst_load, inst_cen,  
                           inst_clk, inst_reset, count_inst );  
  
    parameter width = 8;  
  
    input [width-1 : 0] inst_data;  
    input inst_load;  
    input inst_cen;  
    input inst_clk;  
    input inst_reset;  
    output [width-1 : 0] count_inst;  
  
    // Instance of DW03_lfsr_load  
    DW03_lfsr_load #(width)  
        U1 ( .data(inst_data), .load(inst_load), .cen(inst_cen), .clk(inst_clk),  
            .reset(inst_reset), .count(count_inst) );  
  
endmodule
```

Revision History

For notes about this release, see the [DesignWare Building Block IP Release Notes](#).

For lists of both known and fixed issues for this component, refer to the [STAR report](#).

For a version of this datasheet with visible change bars, click [here](#).

Date	Release	Updates
September 2024	DWBB_202409.0	■ Updated date and version; no technical changes
December 2023	DWBB_202312.0	■ Updated date and version; no technical changes
March 2019	DWBB_201903.0	■ Removed minPower designation from this datasheet
January 2019	DWBB_201806.5	■ Updated example in “ HDL Usage Through Component Instantiation - VHDL ” on page 4 ■ Added this Revision History table and the document links on this page

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